

RACHNA KADAM

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SUMMARY

NYU Tandon M.S. candidate in Mechatronics & Robotics with hands-on experience building mobile robots from scratch using embedded hardware, sensors, and closed-loop control. Worked across the full stack from microcontrollers and motor drivers to ROS 2, computer vision, and path planning.

SKILLS

- **Languages:** Embedded C, Python, C++, MATLAB
- **Embedded Systems:** ESP32, Arduino, Raspberry Pi, Jetson; I2C, UART, SPI, PLC Programming (IndraWorks – Bosch Rexroth), SCADA/HMI
- **Sensors:** LiDAR, IMU, Quadrature/Magnetic Encoders, Current/Voltage, Ultrasonic, IR, Temperature, Monocular/Stereo Camera
- **Controls:** PID, LQR, LSD, System Identification, State Estimation, Model-Based Design, Real-Time Feedback Control
- **Robotics:** ROS 2, Localization, SLAM, Path Planning, Kinematics & Dynamics, Sensor Fusion
- **AI & Perception:** OpenCV, ArUco Detection, Vision-Language Models, LLM Integration
- **Tools & CAD/CAE:** Git/GitHub, Linux (Ubuntu), SolidWorks, Fusion 360, Ansys, 3D Printing, Laser Cutting

EXPERIENCE

Research Assistant - NYU Mechatronics, Controls and Robotics Lab

June 2025 – Present

VLM-LLM Integrated Multi-Robot System for Autonomous Task Execution

- Designed an interrupt-driven, 4-protocol wireless bridge (Wi-Fi, BLE, Classic BT, ESP-NOW) on ESP32, with mode-switching via 5s button hold.
- Built a custom binary encoding layer compressing JSON payloads from 55 bytes → 7 bytes (87% reduction), sustaining throughput across scaled multi-robot deployments.
- Validated across 4 heterogeneous MCU architectures - ATmega328P + HC-06, ESP32-S2, ESP32-S3, RP2040 W - with zero packet loss during multi-robot coordination.
- Integrated bridge into a full autonomy stack - RRT* planning → Pure Pursuit + PID, achieving <50ms command latency from LLM to hardware execution.

Course Assistant – Automatic Controls Laboratory

Sep 2025 – Present

New York University, Tandon School of Engineering

- Conducted controls lab sessions for 50+ students across DC motor position, magnetic levitation, inverted pendulum, and coupled water tank systems using Quanser + MATLAB-Simulink.
- Redesigned 3 embedded control experiments - FOPDT temperature control, SOPDT DC motor speed control, LQR inverted pendulum - reducing setup time by 60% over Quanser-based equivalents.
- Developed a Python GUI for system ID, loop-shaping, Tustin discretization, and real-time visualization, cutting controller implementation time by 30% and flashing final designs onto Arduino R4 WiFi via embedded C.

Mechanical Design & Validation Engineer (SAE BAJA)

Sep 2021 – May 2022

MSPTME Racing Team

- Designed an ATV hydraulic braking system in Fusion360 CAD, achieving 15% weight reduction validated across 40+ dynamic test runs.
- Ran FEA (Ansys) stress analysis with 96% correlation to field results, securing 10th place in SAE BAJA Phase 2.

PROJECTS

Mecanum Drive SLAM Robot

Feb 2026 – April 2026

- Designed a 4-wheel mecanum drive embedded system on ESP32-S3, interfacing 4x AS5600 magnetic encoders, MPU9250 9-axis IMU, and 12V battery monitoring via 12-bit ADC voltage divider, achieving 95% sensor read reliability at 100Hz.
- Implemented a 3-tier ROS2 architecture - ESP32-S3 → Raspberry Pi 3B → laptop - reducing command latency by 35% over single-board alternatives.
- Deployed SLAM Toolbox + Nav2 on RPLidar A1M8 data, fusing wheel odometry + IMU via Complementary + Kalman Filter, improving localization accuracy by 40% over odometry alone, visualized in RViz.
- Deployed Sliding Mode Controller (SMC) for robust per-wheel motor control, reducing velocity tracking error by 28% over standard PID under lateral disturbances.

Interactive Motion-Controlled Rehabilitation Game: Punch n Pop

April 2025 – May 2025

- Developed a multimodal motion-tracking stack using OpenCV HSV filtering and dual MPU6050 IMUs, achieving sub-100ms end-to-end latency.
- Formulated a sensor fusion pipeline on Raspberry Pi 400 that processes 200Hz acceleration data to compute numerical jerk derivatives for 95% detection reliability.
- Streamlined a wireless communication bridge using the ESP-NOW protocol, reducing transmission overhead by 40% compared to standard WiFi.

Pneumatic Actuated Parking System

Oct 2023 – Nov 2023

- Programmed an Arduino Mega 2560 to orchestrate a full sensor-actuator pipeline — IR (MH-B) + 2x HC-SR04 ultrasonic sensors → SG90 servo barricade → 5/2 DCV-controlled double-acting pneumatic cylinder - with precise distance-threshold based actuation logic.
- Engineered relay-based high-voltage switching (220V AC) via 2-channel SRD-05VDC relay module, isolating MCU logic from pneumatic actuator power with zero false triggers during testing.
- Validated end-to-end automation across 3 sequential actuation stages — entry detection, cylinder extension, cylinder retraction — with <500ms sensor-to-actuator response time.

EDUCATION

New York University, New York, NY

Master of Science, Mechatronics and Robotics

May 2026

GPA: 3.61

NMIMS University, Mumbai, India

Bachelor of Technology, Mechatronics

May 2024

GPA: 3.69